Effects of sorbic acid and its salts on chromosome aberrations, sister chromatid exchanges and gene mutations in cultured Chinese hamster cells.

The ability of sorbic acid and its potassium and sodium salts to induce chromosome aberrations, sister chromatid exchanges (SCE) and gene mutations in cultured Chinese hamster V79 cells was examined. Sodium sorbate caused significant induction of chromosome aberrations and SCE, and also induced 6-thioguanine-resistant mutations in a dose-dependent manner. The clastogenic potency of sodium sorbate was found to be less than one hundredth of that of the potent clastogen N-methyl-N'-nitro-N-nitrosoguanidine. The induction of SCE by sodium sorbate was twice the control level, whereas that by methyl methanesulphonate, a potent inducer of SCE, was 14 times the control level. The mutagenic potency of sodium sorbate was less than one-tenth that of ethyl methanesulphonate, a potent inducer of mutation, when compared at an equitoxic level. Sorbic acid and its potassium salt induced chromosome aberrations, but only at the highest doses tested. These compounds also induced 1.2 times the control level of SCE, but neither compound induced 6-thioguanine-resistant mutations. The cytogenetic activity of sodium sorbate was concluded not to be due to the effect of osmotic pressure or an impurity. These results indicate that sodium sorbate is a genotoxic agent, although its potency seems to be weak, and that sorbic acid and potassium sorbate are less genotoxic than the sodium salt.